
LETERO (Letrozole Tablets 2.5 mg)

NAME OF THE PRODUCT

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NAME AND STRENGTH OF ACTIVE INGREDIENT

Name : Letrozole

Strength : 2.5 mg

PRODUCT DESCRIPTION

Letrozole Tablets 2.5 mg

Dark yellow colored, round shaped, slightly biconvex bevel edged film coated tablets debossed with '5' on one side and 'H' on the other side.

PHARMACODYNAMICS

Pharmacotherapeutic group: Endocrine therapy. Hormone antagonist and related agents: aromatase inhibitor, ATC code: L02BG04.

Pharmacodynamic effects

The elimination of oestrogen-mediated growth stimulation is a prerequisite for tumour response in cases where the growth of tumour tissue depends on the presence of oestrogens and endocrine therapy is used. In postmenopausal women, oestrogens are mainly derived from the action of the aromatase enzyme, which converts adrenal androgens - primarily androstenedione and testosterone - to oestrone and oestradiol. The suppression of oestrogen biosynthesis in peripheral tissues and the cancer tissue itself can therefore be achieved by specifically inhibiting the aromatase enzyme.

Letrozole is a non-steroidal aromatase inhibitor. It inhibits the aromatase enzyme by competitively binding to the haem of the aromatase cytochrome P450, resulting in a reduction of oestrogen biosynthesis in all tissues where present.

Letrozole is highly specific in inhibiting aromatase activity. Impairment of adrenal steroidogenesis has not been observed. No clinically relevant changes were found in the plasma concentrations of cortisol, aldosterone, 11-deoxycortisol, 17-hydroxyprogesterone, and ACTH or in plasma renin activity. Glucocorticoid and mineralocorticoid supplementation is not necessary.

No changes were noted in plasma concentrations of androgens (androstenedione and testosterone) indicating that the blockade of oestrogen biosynthesis does not lead to accumulation of androgenic precursors. Plasma levels of LH and FSH are not affected by letrozole in patients, nor is thyroid function as evaluated by TSH, T4, and T3 uptake test.

PHARMACOKINETICS

Absorption

Letrozole is rapidly and completely absorbed from the gastrointestinal tract (mean absolute bioavailability: 99.9%). It was reported that food slightly decreases the rate of absorption (median tmax 1 hour fasted versus 2 hours fed; and mean Cmax 129 ± 20.3 nmol/litre fasted versus 98.7 ± 18.6 nmol/litre fed) but the extent of absorption (AUC) is not changed. The minor effect on the absorption rate is not considered to be of clinical relevance, and therefore letrozole may be taken without regard to mealtimes.

Distribution

Plasma protein binding of letrozole is approximately 60%, mainly albumin (55%). The concentration of letrozole in erythrocytes is about 80% of that in plasma. It was reported that after administration of 2.5 mg ¹⁴C-labelled letrozole, approximately 82% of the radioactivity in plasma was unchanged compound. Systemic exposure to metabolites is therefore low. Letrozole is rapidly and extensively distributed to tissues. Its apparent volume of distribution at steady state is about 1.87 ± 0.47 l/kg.

Biotransformation

Metabolic clearance to a pharmacologically inactive carbinol metabolite is the major elimination pathway of letrozole (CL_m = 2.1 l/h) but is relatively slow when compared to hepatic blood flow (about 90 l/h). The cytochrome P450 isoenzymes 3A4 and 2A6 were reported to be capable of converting letrozole to this metabolite. Formation of minor unidentified metabolites and direct renal and faecal excretion play only a minor role in the overall elimination of letrozole. It was reported that within 2 weeks after administration of 2.5 mg ¹⁴C-labelled letrozole to healthy postmenopausal volunteers, $88.2 \pm 7.6\%$ of the radioactivity was recovered in urine and $3.8 \pm 0.9\%$ in faeces. At least 75% of the radioactivity recovered in urine up to 216 hours ($84.7 \pm 7.8\%$ of the dose) was attributed to the glucuronide of the carbinol metabolite, about 9% to two unidentified metabolites, and 6% to unchanged letrozole.

Elimination

The apparent terminal elimination half-life in plasma is about 2 to 4 days. After daily administration of 2.5 mg steady-state levels are reached within 2 to 6 weeks. Plasma concentrations at steady state are approximately 7 times higher than concentrations measured after a single dose of 2.5 mg, while they are 1.5 to 2 times higher than the steady-state values predicted from the concentrations measured after a single dose, indicating a slight non-linearity in the pharmacokinetics of letrozole upon daily administration of 2.5 mg. Since steady-state levels are maintained over time, it can be concluded that no continuous accumulation of letrozole occurs.

Linearity/non-linearity

The pharmacokinetics of letrozole was reported to be dose proportional after single oral doses up to 10 mg (dose range: 0.01 to 30 mg) and after daily doses up to 1.0 mg (dose range: 0.1 to 5mg). After a 30 mg single oral dose there was a slightly dose over- proportional increase in AUC value. The dose over-proportionality is likely to be the result of a saturation of metabolic elimination processes. Steady levels were reached after 1 to 2 months at all dosage regimens tested (0.1-5.0 mg daily).

Special populations

Elderly

Age had no effect on the pharmacokinetics of letrozole.

Renal impairment

No effect on the pharmacokinetics of letrozole was reported after a single dose of 2.5 mg. Therefore, no dose adjustment is required for patients with renal impairment (CL_{cr} ≥ 10 mL/min). Little information is available in patients with severe impairment of renal function (CL_{cr} <10 mL/min).

Hepatic impairment

Letrozole should be administered with caution to patients with severe hepatic impairment and after consideration of the risk/benefit in the individual patient.

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INDICATIONS

- Adjuvant treatment of postmenopausal women with hormone receptor positive early breast cancer.
- Adjuvant treatment of postmenopausal women with early breast cancer (positive or unknown oestrogen or progesterone receptor status) who have received 5 years of adjuvant tamoxifen therapy (extended adjuvant therapy).
- First-line treatment in postmenopausal women with hormone-dependent advanced breast cancer.
- Treatment of advanced breast cancer in women with natural or artificially induced postmenopausal status, who have previously been treated with antioestrogens.
- Pre-operative therapy in postmenopausal women with localized hormone receptor positive breast cancer, to allow subsequent breast-conserving surgery in women not originally considered candidates for this type of surgery. Subsequent treatment after surgery should be in accordance with standard of care.

RECOMMENDED DOSE

Adults

The recommended dose of Letrozole is 2.5 mg once daily. In the adjuvant and extended adjuvant setting, treatment with Letrozole should continue for 5 years or until disease relapse/recurrence occurs, whichever comes first. In the large pivotal study of letrozole versus tamoxifen in the adjuvant setting, no benefit in efficacy or safety was obtained by sequential administration of these treatments compared with letrozole administered continuously for 5 years. In patients with metastatic disease, treatment with Letrozole should continue until tumor progression is evident. In the pre-operative setting, treatment with Letrozole should be continued for 4 to 8 months in order to establish optimal tumor reduction. If the response is not adequate, treatment with Letrozole should be discontinued, surgery scheduled and/or further treatment options discussed with the patient.

SPECIAL POPULATIONS

Hepatic impairment

No dose adjustment of Letrozole is required for patients with mild to moderate hepatic insufficiency (Child-Pugh score A or B). Insufficient data are available for patients with severe hepatic impairment, but patients with severe hepatic impairment (Child-Pugh score C) should be kept under close supervision.

Renal impairment

No dosage adjustment of Letrozole is required for patients with renal insufficiency with (creatinine clearance (CLcr $\geq$ 10 mL/min.) Insufficient data are available in cases of renal insufficiency with CLcr <10 mL/min.

Pediatrics

Letrozole is not recommended for use in children and adolescents. The safety and efficacy of Letrozole in children and adolescents aged up to 17 years have not been established. Limited data are available and no recommendation on a posology can be made.

Geriatrics

No dose adjustment is required for elderly patients.

MODE OF ADMINISTRATION

Oral - The tablet can be taken with or without food and should be swallowed whole with a glass of water or another liquid.

CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients is described in lateral sections.

- Premenopausal endocrine status
- Pregnancy
- Breast-feeding

WARNINGS AND PRECAUTIONS

Menopausal status

In patients whose menopausal status is unclear, luteinising hormone (LH), follicle- stimulating hormone (FSH) and/or oestradiol levels should be measured before initiating treatment with Letrozole Tablets. Only women of postmenopausal endocrine status should receive Letrozole Tablets.

Renal impairment

Letrozole Tablets has not been investigated in a sufficient number of patients with a creatinine clearance lower than 10 mL/min. The potential risk/benefit to such patients should be carefully considered before administration of Letrozole Tablets.

Hepatic impairment

In patients with severe hepatic impairment (Child-Pugh C), systemic exposure and terminal half-life were approximately doubled compared to healthy volunteers. Such patients should therefore be kept under close supervision.

Bone effects

Letrozole Tablets is a potent oestrogen-lowering agent. Women with a history of osteoporosis and/or fractures, or who are at increased risk of osteoporosis, should have their bone mineral density formally assessed prior to the commencement of adjuvant and extended adjuvant treatment and monitored during and following treatment with letrozole. Treatment or prophylaxis for osteoporosis should be initiated as appropriate and carefully monitored. In the adjuvant setting a sequential treatment schedule (letrozole 2 years followed by tamoxifen 3 years) could also be considered depending on the patient's safety profile.

Other warnings

Co-administration of Letrozole Tablets with tamoxifen, other anti-oestrogens or oestrogen- containing therapies should be avoided as these substances may diminish the pharmacological action of letrozole.

As the tablets contain lactose, Letrozole Tablets is not recommended for patients with rare hereditary problems of galactose intolerance, of severe lactase deficiency or of glucose- galactose malabsorption.

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INTERACTION WITH OTHER MEDICAMENTS

There is no clinically significant drug interactions from coadministration of letrozole with cimetidine and warfarin .

There is no evidence of clinically relevant interactions with other commonly prescribed drugs. There is no clinical experience on the use of letrozole in combination with other anticancer agents.

Letrozole inhibits in vitro the cytochrome P450-isozymes 2A6, and moderately 2C19. CYP2A6 does not play a major role in drug metabolism. Thus, clinically relevant interactions with CYP2C19 are unlikely to occur. However, caution should be used in the concomitant administration of drugs whose disposition is mainly dependent on these isoenzymes and whose therapeutic index is narrow.

Concurrent use of letrozole and tamoxifen may result in reduced letrozole serum concentrations.

PREGNANCY AND LACTATION

Women of perimenopausal status or child-bearing potential

Letrozole Tablets should only be used in women with a clearly established postmenopausal status. As there are reports of women regaining ovarian function during treatment with Letrozole Tablets despite a clear postmenopausal status at start of therapy, the physician needs to discuss adequate contraception when necessary.

Pregnancy

Based on human experience in which there have been isolated cases of birth defects (labial fusion, ambiguous genitalia), Letrozole Tablets may cause congenital malformations when administered during pregnancy. Studies in animals have shown reproductive toxicity.

Letrozole Tablets is contraindicated during pregnancy.

Breast-feeding

It is unknown whether letrozole and its metabolites are excreted in human milk. A risk to the newborns/infants cannot be excluded.

Letrozole Tablets is contraindicated during breast-feeding.

Fertility

The pharmacological action of letrozole is to reduce oestrogen production by aromatase inhibition. In premenopausal women, the inhibition of oestrogen synthesis leads to feedback increases in gonadotropin (LH, FSH) levels. Increased FSH levels in turn stimulate follicular growth and can induce ovulation.

Side effects

The most frequently reported adverse reactions were hot flushes, hypercholesterolaemia, arthralgia, fatigue, increased sweating and nausea.

Important additional adverse reactions that may occur are: skeletal events such as osteoporosis and/or bone fractures and cardiovascular events (including cerebrovascular and thromboembolic events). The frequency category for these adverse reactions is described in Table 1

Table 1: Tabulated listing of adverse reactions

Infections and infestations	
Uncommon:	Urinary tract infection
Neoplasms, benign, malignant and unspecified (including cysts and polyps)	
Uncommon:	Tumour pain ¹
Blood and the lymphatic system disorders	
Uncommon:	Leukopenia
Immune system disorders	
Not known:	Anaphylactic reaction
Metabolism and nutrition disorders	
Very common:	Hypercholesterolaemia
Common:	Anorexia, appetite increase
Psychiatric disorders	
Common:	Depression
Uncommon:	Anxiety (including nervousness), irritability
Nervous system disorders	
Common:	Headache, dizziness
Uncommon:	Somnolence, insomnia, memory impairment, dysaesthesia (including paraesthesia, hypoaesthesia), taste disturbance, cerebrovascular accident, carpal tunnel syndrome

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Eye disorders	
Uncommon	Cataract, eye irritation, blurred vision
Cardiac disorders	
Uncommon:	Palpitations ¹ , tachycardia, ischaemic cardiac events (including new or worsening angina, angina requiring surgery, myocardial infarction and myocardial ischaemia)
Vascular disorders	
Very common:	Hot flushes
Common:	Hypertension
Uncommon:	Thrombophlebitis (including superficial and deep vein thrombophlebitis)
Rare:	Pulmonary embolism, arterial thrombosis, cerebrovascular infarction
Respiratory, thoracic and mediastinal disorders	
Uncommon:	Dyspnoea, cough
Gastrointestinal disorders	
Common:	Nausea, dyspepsia ¹ , constipation, abdominal pain, diarrhoea, vomiting
Uncommon:	Dry mouth, stomatitis ¹
Hepatobiliary disorders	
Uncommon:	Increased hepatic enzymes
Not known:	Hepatitis
Skin and subcutaneous tissue disorders	
Very common:	Increased sweating
Common:	Alopecia, rash (including erythematous, maculopapular, psoriaform, and vesicular rash), dry skin
Uncommon:	Pruritus, urticaria
Not known:	Angioedema, toxic epidermal necrolysis, erythema multiforme
Musculoskeletal and connective tissue disorders	
Very common:	Arthralgia
Common:	Myalgia, bone pain ¹ , osteoporosis, bone fractures
Uncommon:	Arthritis
Not known:	Trigger finger
Renal and urinary disorders	
Uncommon:	Increased urinary frequency
Reproductive system and breast disorders	
Common:	Vaginal bleeding
Uncommon:	Vaginal discharge, vaginal dryness, breast pain
General disorders and administration site conditions	
Very common:	Fatigue (including asthenia, malaise)
Common:	Peripheral oedema
Uncommon:	General oedema, mucosal dryness, thirst, pyrexia

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Investigations	
Common:	Weight increase
Uncommon:	Weight loss

¹ Adverse drug reactions reported only in the metastatic setting

Description of selected adverse reactions

Cardiac adverse reactions

In the adjuvant setting, the following adverse reactions had been reported: angina requiring surgery; cardiac failure; hypertension; cerebrovascular accident/transient ischaemic attack. In the extended adjuvant setting, the following adverse reactions were reported: angina requiring surgery; new or worsening angina; myocardial infarction; thromboembolic event; stroke/transient ischaemic attack.

Skeletal adverse reactions

In the extended adjuvant setting, it was reported that bone fractures or osteoporosis were reported to occur.

SYMPTOMS AND TREATMENT OF OVERDOSE

No specific treatment for overdose is known; treatment should be symptomatic and supportive.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

This medicine may affect your ability to drive or use machines. If the tablets make you feel sick, dizzy or tired, or give you a headache, do not drive or use machines and contact your doctor immediately.

Letero contains lactose (milk sugar). If you have been told by your doctor that you have an intolerance to some sugars, contact your doctor before taking this medicine.

STORAGE CONDITION

Store below 30°C and protect from moisture.

DOSAGE FORMS AND PACKAGING AVAILABLE

10 x 10's tablets; Blister pack

PRODUCT REGISTRATION HOLDER

Camber Laboratories Sdn Bhd

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MANUFACTURER

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DATE OF REVISION

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